Holistic Mutual Representation Enhancement for Few-Shot Remote Sensing Segmentation

Yuyu Jia, Junyu Gao, *Member, IEEE*, Wei Huang, Yuan Yuan, *Senior Member, IEEE*, and Qi Wang, *Senior Member, IEEE*

***Abstract*— Few-shot segmentation (FSS) endeavors to utilize a minimal amount of annotated samples (*support*) to guide the segmentation of unseen objects (*query*). Previous techniques primarily employ a *support-to-query* paradigm, neglecting to sufficiently leverage the mutual representation between query and support images, which leaves models suffering from intra-class variations and background interference in remote sensing images. This article proposes a holistic mutual representation enhancement (HMRE) method to bridge these gaps. First, a dual activation (DA) module is devised to establish information symmetry between the two branches and forms the foundation for mutual representation enhancement. Subsequently, the holistic mutual enhancement is jointly constructed by the global semantic (GS) and spatial dense (SD) mutual enhancement modules. In the prediction stage for segmentation, we integrate the enhanced mutual representation into the mutual-fusion decoder to activate the homologous object regions bidirectionally. To expedite the replication of investigation in this task, we further create a corresponding benchmark Flood-3i. The whole dataset is attainable at https://drive.google.com/drive/folders/1FMAKf2sszoFKjq0UrUmSLnJDbwQSpfxR. Extensive experiments on two benchmarks iSAID-5i and Flood-3i demonstrate the superiority of our proposed method, which also sets a new state-of-the-art.**

***Index Terms*— Few-shot semantic segmentation, mutual representation enhancement, remote sensing images.**

I. Introduction

SEMANTIC segmentation plays a pivotal role in the domain of remote sensing image interpretation. It has been widely explored for practical applications, such as building/road extraction [1], land-cover mapping [2], [3], disaster assessment [4], and environmental protection [5]. Recently, deep learning techniques [6], [7], [8], [9] have greatly contributed to the remarkable advancements in remote sensing image segmentation algorithms. However, these successful deep-learning techniques intensely depend on extensive pixel-level annotations, which incur substantial time and labor costs for practical applications. Moreover, their generalization capability to novel tasks could be fragile [10].

In that case, few-shot learning (FSL) has been extensively investigated. It acquires transferable knowledge and subsequently generalizes it to novel (unseen) tasks with only a few labeled samples. Likewise, image segmentation inherits the paradigm of FSL, constructing a meaningful and challenging task, termed few-shot segmentation (FSS). Given an image (*query*) to be segmented in the test set, the FSS model utilizes the information from several densely annotated samples (*support*) to identify target regions containing the predefined semantic category. Current FSS methods normally embrace a two-branch structure to separately extract the support and query feature, in which the support branch widely uses the mask average pooling [11] operation to acquire class-specific features related to the target region. Then, the class-specific cues are embedded into the query branch to guide the segmentation through pixel-wise similarities. These techniques unidirectionally transmit category representation (*support-to-query*), formulating the FSS task as a guided segmentation task. However, under such a paradigm, the mutual information between query and support remains underutilized in addressing intra-class variances presented by objects of the same category with diverse appearances [see Fig. 1(b)]. Intuitively, the query information could, in turn, direct the segmentation of the support (*query-to-support*).

Taking these into consideration, some research in natural image processing employs symmetric structures for two-branch interaction. PANet [12] introduces a prototype alignment strategy that performs bidirectional prediction between support and query. BriNet [13] reports an information exchange module to learn the nonlocal spatial transforms between support and query images, which is more complex and omits the global representation in each channel. CRNet [14] proposes a cross-reference module to enhance the global mutual representation in the two branches. Despite the recent progress showcased by these investigations, they still struggle with the following points.

1) The enhancement of global mutual representation is hasty, as the incorporation of global average pooling inevitably introduces a large amount of background noise. As shown in Fig. 1(a), due to the wide geographical coverage of remote sensing images, nontarget regions

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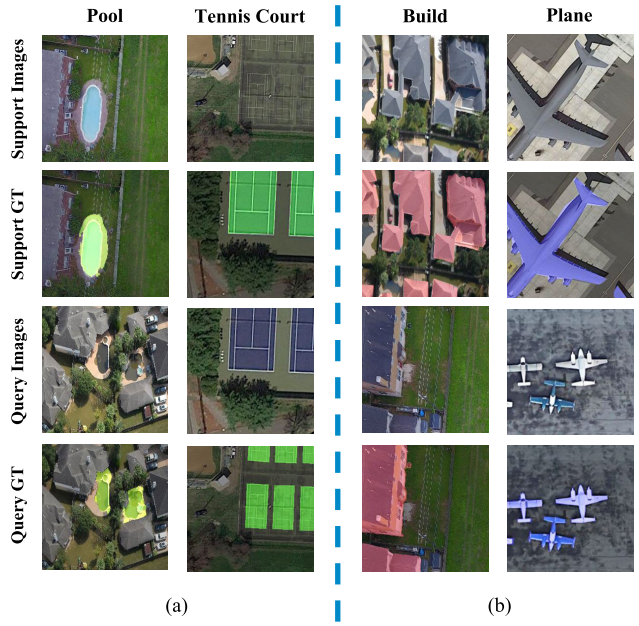


Fig. 1. Demonstration of the issues in handling the FSS task for remote sensing images. (a) Non-target regions with the same category in the background of both support and query images (e.g., buildings or trees) could interfere with the segmentation of target regions (e.g., pool). (b) Variations in appearance, size, and distribution of objects (e.g., planes) belonging to the same category in the support and query images are substantial.

of the same category often appear concurrently in the background of both support and query samples, which may mislead target region segmentation.

2) The enhancement of spatial mutual representation with only mid-level features is insufficient for capturing target location. It overlooks the valuable integration of high-level features, which provide complementary information necessary for precise localization.
3) The absence of a holistic framework that concurrently investigates both global semantic (GS) mutual enhancement and spatially dense mutual enhancement of representations.

In this work, we develop a holistic mutual representation enhancement (HMRE) method for overcoming the constraints of existing works. Fig. 2 depicts the flow of data in HMRE and concisely showcases the role of each component. Given a pair of support and query images, along with a support mask, a proposed dual activation (DA) module activates a pseudo-mask for the query image and a prior map for the support image, respectively. This process achieves information symmetrization of the two branches, laying the foundation for mutual representation enhancement. Subsequently in a GS mutual enhancement module, the generated pseudo-mask is utilized to extract a pure target region prototype in the query image. Then, it combines the support prototype to compute a coefficient for GS mutual enhancement. Intuitively, only the channels sharing analogous global target semantics in both branches will receive enhancement through the aforementioned coefficient. In addition to this, we also devise a spatial dense (SD) mutual enhancement module to generate the SD coefficient for each branch. Compared with previous spatial interaction strategies, the proposed approach incorporates high-level spatial information (i.e., the support and query prior maps), which offers complementary insights essential for accurate localization. Finally, with the two types of coefficients, the mutual fusion decoder achieves holistic enhancement of mutual representation and optimizes the above processes using bidirectional prediction.

Moreover, there is a shortage of remote sensing datasets specifically tailored for the FSS. To our best knowledge, only two datasets (i.e., iSAID-5i [15], dense labeling remote sensing dataset (DLRSD) [16]) have been used for this purpose, where the scenes have simple backgrounds or few target objects. Considering this fact, we create a new benchmark constructed from FloodNet [17], named Flood-3i. Compared to DLRSD, the proposed dataset offers more samples, allowing for a comprehensive evaluation of algorithms. Meanwhile, each image in this dataset has a background with a wider variety of object categories than iSAID-5i, providing more challenges for the FSS task. Further details regarding this dataset will be elaborated in Section IV.

Our main contributions can be outlined as follows.

1) The GS mutual enhancement module is designed to mitigate the interference from nontarget regions in the background while enhancing the GSs of co-occurring target regions in the two branches.
2) The proposed SD mutual enhancement module lightly fuses the positional information of high-level feature maps and densely enhances the spatial representation of the two branches.
3) A new benchmark, Flood-3i, is meticulously created for the FSS of remote sensing images. The HMRE method proposed in this study is evaluated on two benchmarks (i.e., Flood-3i and iSAID-5i), where it outperforms the prior leading techniques.

II. Related Works

A. Semantic Segmentation

Semantic segmentation assigns predefined semantic categories to each pixel, which is one of the most foundational image-processing tasks the fully convolutional network (FCN)-based models [18], [19] substitute fully connected (FC) layers with convolutional layers to preserve the spatial information. Furthermore, a series of optimization algorithms improve the above model structures, such as dilated convolutions [20], pyramid pooling [21], atrous spatial pyramid pooling (ASPP) [22], attention mechanism [23], encoder–decoder [24], multiscale feature aggregation [25], etc.

As for the interpretation of remote sensing images, deep learning-based semantic segmentation technologies also attract widespread attention from researchers [26], [27], [28]. For instance, He et al. [29] proposes MANet for extracting perceptive and multiscale representations in remote sensing images. Chen et al. [30] proposes AERFC, which adaptively controls convolution sampling locations and adjusts effective receptive fields to reduce training difficulty and preserve image details. HFGNet [26] places emphasis on the significance of background information and enriches the foreground saliency features.

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Despite achieving high segmentation accuracy, these methods are limited by the requirement for a substantial amount of annotated data. Furthermore, they often encounter difficulties when facing unseen target objects. To effectively alleviate these, we undertake the FSS task and fully explore a class-agnostic mutual representation enhancement method to boost the model's generalization.

B. Few-Shot Learning

FSL aims to generalize knowledge to novel categories by leveraging a small amount of annotated data [33]. Currently, the majority of FSL algorithms are situated within the meta-learning framework. Generally speaking, these studies can be grouped into two branches: metric-based approaches [34], [35], [36], [37] and optimization-based approaches [38], [39], [40], [41]. Metric-based methods categorize a query sample according to its relative distance to each category prototype in the feature space. Optimization-based methods focus on learning optimal initialization parameters or optimizers that allow quick adaptation to novel classes.

Acquiring a vast collection of annotated remote sensing images presents a formidable challenge, rendering the FSL in this domain a task that is both significant and arduous. Researchers have dedicated extensive efforts to this undertaking. For example, Li et al. [42] presents DLA-MatchNet, which is designed via discriminative learning of adaptive matching. Li et al. [43] proposes SCL-MLNet to address the limitations of existing methods in utilizing limited annotated data and handling intra-class sample variations. Jia et al. [44] devises a multiview-attention mechanism to explore the information across different views of images, and boosts the model's generalization by optimizing feature distributions of hard samples. To capture intrinsic characteristics across semantic classes during the training of all base classes, Ji et al. [45] employs a cross-entropy loss with two auxiliary objectives.

Previous studies on FSL have primarily concentrated on accomplishing classification tasks. In contrast, our research tackles the more demanding task of few-shot semantic segmentation, which contributes to achieving a more sophisticated comprehension of remote sensing images.

C. Few-Shot Segmentation

FSS inherits the main settings of FSL and achieves dense prediction of images, which is first proposed by Shaban et al. [46], termed OSLSM. Similar to OSLM, the majority of subsequent works have adopted a dual-branch network structure (i.e., a support branch and a query branch). Specifically, Shaban et al. [46] first utilize a mask average pooling approach to gather guidance information that exclusively considers pixels belonging to the support image. Zhang et al. [47] introduces CANet, a novel segmentation network that leverages a two-branch dense comparison module and an iterative optimization module for performing multi-level feature comparison and refining predicted results. Siam and Oreshkin [48] presents an adaptive masked proxies strategy, which leverages multiresolution average pooling on masked embeddings to construct the final segmentation layer weights for new classes. Later, PFENet [49] introduces the training-free prior mask, which significantly improves the model's generalization performance on unseen categories. On this basis, some relevant works further improve the model and achieve reliable performance, such as SD-AANet [50], MMNet [51], SCLNet [52], and PFENet++ [53].

Specific to the research of FSS of the remote sensing image, there is currently limited work in this area, therefore further investigations are needed. In current employment, Wang et al. [16] develops an affinity-based fusion mechanism to handle the intra-class variations. Yao et al. [15] mitigates the substantial variance in object appearances through a detailed matching module and a scale-aware focal loss, and more importantly, they provided a benchmark iSAID-5i for the research of FSS in remote sensing images. Both these two methods address the crucial challenge of the FSS task and propose corresponding solutions from the view of *support to query*. Instead, this study focuses on enhancing the two-branch mutual representation and activating the homologous object regions in a bidirectional manner.

III. Methodology

A. Task Description

FSS is the task of segmenting objects in a query image leveraging only a limited number of labeled samples. To be more precise, the model is trained on the training set $\mathcal{D}_{\text{train}}$ and its performance is evaluated on the testing set $\mathcal{D}_{\text{test}}$. Assuming that $\mathcal{C}_{\text{train}}$ and $\mathcal{C}_{\text{test}}$ correspond to the sets of categories $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{test}}$, respectively. These two sets of categories have no overlapping elements, i.e., $\mathcal{C}_{\text{train}} \cap \mathcal{C}_{\text{test}}$. We adopt the episodic learning strategy for training as prior studies in FSS [54], [55].

For a K-shot segmentation setting, each episode contains: 1) a support set $\mathcal{S} = \{(I_s^k, M_s^k)\}_{i=1}^K$, where I_s^k is the kth support image and M_s^k is the corresponding binary mask; and 2) a query set $\mathcal{Q} = \{(I_q, M_q)\}$, where I_q denotes the query image and M_q represent the corresponding binary mask. The support set $\mathcal{S}$ and the query set $\mathcal{Q}$ belong to the same category space. During the training step, both the query mask M_q and the support masks M_s are utilized, whereas only the support masks M_s are available during testing. Since the learned model is inferred without further optimization for novel categories during testing, it is crucial to train the model to find the co-occurred target regions between support images and the query image.

B. Method Overview

To alleviate intra-class variances and background interference in remote sensing images, we propose the HMRE strategy. To facilitate the description, we illustrate the pipeline of HMRE under the one-shot scenario in Fig. 2. It contains three constituents (i.e., DA module, GS, and SD mutual enhancement modules) and a mutual-fusion decoder.

To elaborate, we adopt the pretrained backbones (ResNet50 [31] or VGG16 [32]) with shared weights to extract both mid- and high-level features. Given the support mask M_s and the query prior mask produced by the prior generation

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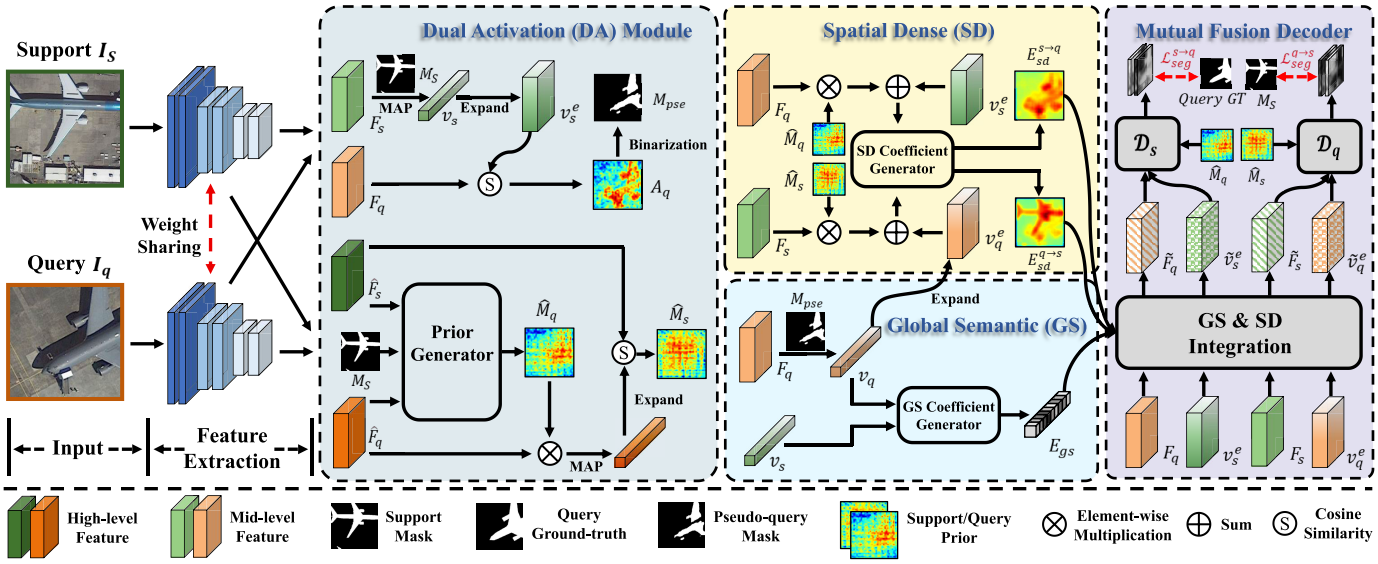


Fig. 2. Pipeline of the HMRE method for one-shot setting. Given a query image, a support image, and a support mask, the pretrained ResNet50 [31] or VGG16 [32] with fixed weights is leveraged to extract mid-level and high-level features. To achieve symmetry of the information between the two branches, we complete a pseudo query mask and a support prior mask in the DA module. Based on this, we devise the GS and SD mutual enhancement modules to calculate corresponding coefficients. Finally, the mutual fusion decoder instantiates the HMRE by weighting the prototypes and mid-level features. Moreover, we adopt a bidirectional prediction paradigm to optimize the above process.

[49], we have the DA whose task is to further activate the support prior mask and pseudo query mask. We then decompose HMRE as GS and SD mutual enhancement modules. GS leverages the pseudo query mask and support mask to squeeze mid-level feature maps into pure prototypes for the two branches, which are used to generate a GS mutual enhancement coefficient. After GS, the SD follows, which fuses the high-level information (i.e., prior masks) and prototypes of the opposite branch, generating the SD mutual enhancement coefficients. Afterward, these two types of mutual enhancement coefficients are fed into the mutual fusion decoder to implement the holistic mutual enhancement. Finally, we achieve optimization of the above process through bidirectional prediction.

C. DA Module

Following the paradigm of leading FSS techniques, we perform the mask average pooling operation on mid-level features to acquire *support prototypes*. As for high-level features, we utilize the prior generator [49] to acquire a *query prior mask* in a training-free manner to maintain generalization ability. Subsequently, these mid- and high-level features are merged and fed into the segmentation decoder, resulting in the generation of predictive masks. However, such an asymmetric dual-branch structure cannot serve as a prerequisite for mutual enhancement. That is, we need to complete the query prototype and the support prior mask.

1) Activation of the Pseudo Query Mask: Inspired by a common ideology that features of the same object exhibit greater similarity compared to those belonging to different objects of the same category [56], we activate the pseudo mask for the query image. This process is constructed on mid-level features. As in [47], we concatenate the features extracted from *block2* and *block3* in ResNet50 or VGG16. Subsequently, a 1×1 *convolution* is adopted to reduce the dimensions of the channels to 256 and formulate mid-level features. Formally, let $F_s \in \mathbb{R}^{C\times H\times W}$ and $F_q \in \mathbb{R}^{C\times H\times W}$ be the mid-level support and query features, where C, H, and W are channel, height, and width, respectively.

Given F_s, F_q, and the support mask M_s, we apply the mask average pooling to calculate the support prototype as follows:

$$v_s = \mathcal{A}_{\text{pool}}(F_s \odot \mathcal{R}(M_s)) \in \mathbb{R}^C \tag{1}$$

where $\mathcal{A}_{\text{pool}}(\cdot)$ is the average-pooling function, $\odot$ represents Hadamard product, and $\mathcal{R}(\cdot)$ reshapes the mask M_s to fit the dimension of the support feature F_s. Afterward, we evaluate the cosine similarity between the support prototype v_s and the query feature F_q at each spatial location, respectively,

$$A_q^{(x,y)} = \frac{v_s^{\mathrm{T}} \cdot F_q^{(x,y)}}{\|v_s\| \cdot \left\|F_q^{(x,y)}\right\|} \tag{2}$$

where $A_q \in \mathbb{R}^{H\times W}$ denotes the activation map of the query feature, and (x, y) indexes its spatial location. In essence, the higher the value of a certain position in the activation image, the more likely it is to be the target region of the query map. Based on this, we use the average value of A_q as a threshold to adaptively generate a pseudo query mask $M_{\text{pse}} \in \{0, 1\}^{H\times W}$ as

$$M_{\text{pse}}^{(x,y)} = \mathbb{I}\left[A_q^{(x,y)} > \mathcal{A}_{\text{avg}}(A_q)\right] \tag{3}$$

where $\mathcal{A}_{\text{avg}}(\cdot)$ stands for the averaging function, and $\mathbb{I}[\cdot]$ is the indicator function.

2) Activation of the Support Prior Mask: Following [49], we convert the pretrained high-level features (i.e., the output derived from the last layer of the backbone) into a class-agnostic prior mask that simply denotes the rough location of target regions. Formally, suppose $\hat{F}_s \in \mathbb{R}^{C\times H\times W}$ and

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$\hat{F}_q \in \mathbb{R}^{C\times H\times W}$ are the high-level support and query features, respectively, the prior mask of the query image can be written as

$$\hat{M}_q = \mathcal{G}(M_s, \hat{F}_s, \hat{F}_q) \tag{4}$$

where $\hat{M}_q \in \mathbb{R}^{H\times W}$, and $\mathcal{G}(\cdot,\cdot,\cdot)$ is the prior generator. For more details please refer to [49].

To further achieve information symmetry between the two branches, we also need to fill in the support prior mask. Note that although the explicit support mask is available, directly introducing the ground truth (GT) of the support sample might not be conducive to optimizing the model from the query branch to the support branch. Therefore, we activate the support prior mask based on $\hat{M}_q$. Specifically, the mask average pooling is conducted on the high-level query feature and its corresponding prior mask

$$\hat{v}_q = \mathcal{A}_{\text{pool}}(\hat{F}_q \odot \hat{M}_q) \tag{5}$$

where $\hat{v}_q \in \mathbb{R}^C$ encodes a proxy of the high-level query feature. Then, we compute the similarity between the high-level support feature $\hat{F}_s$ and the above proxy to produce an activation map

$$\hat{M}_s^{(x,y)} = \frac{\hat{v}_q^{\mathrm{T}} \cdot \hat{F}_s^{(x,y)}}{\left\|\hat{v}_q\right\| \cdot \left\|\hat{F}_s^{(x,y)}\right\|} \tag{6}$$

where $\hat{M}_s$ merely indicates the probability of pixels belonging to the target category in the support feature. Thus, we similarly regard it as a prior mask of the support image.

D. GS Mutual Enhancement Module

In the previous step, we build the information symmetry between the two branches. On this basis, the mutual enhancement of GS representation is subsequently performed. The study in [14] we consider most relevant to our work highlights channels with similar GSs through a cross-reference module. Nonetheless, the utilization of global average pooling may introduce substantial background interference. In cases where common category regions exist in the backgrounds of both the support and query images, this operation might inadvertently lead the model to enhance the GSs of the nontarget channel. To counter this, we opt to use the prototype of the target region as a purer input instead of resorting to the conventional global average pooling method, which simply yet effectively focuses on co-occurrent target regions.

Similarly, we can activate the query prototype leveraging the pseudo query mask generated from the DA module as (1)

$$v_q = \mathcal{A}_{\text{pool}}(F_q \odot M_{\text{pse}}) \in \mathbb{R}^C. \tag{7}$$

As depicted in Fig. 3, the support and query prototypes are each directed to two-layer FC layers followed by a sigmoid activation function, converting the vector into the importance of each channel, confined within the interval [0, 1]. Subsequently, the vectors from the two branches undergo fusion, forming the GS mutual enhancement coefficient E_{gs} through element-wise multiplication as

$$E_{\text{gs}} = \phi_s(v_s) \odot \phi_q(v_q) \in \mathbb{R}^C \tag{8}$$

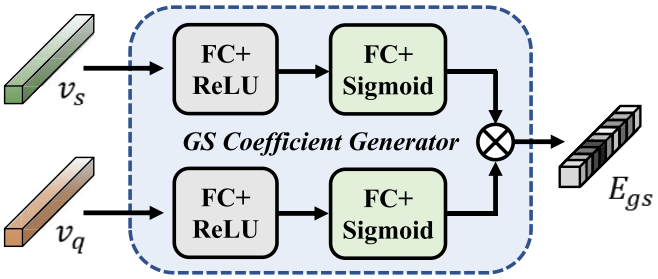


Fig. 3. Illustration of the GS coefficient generator.

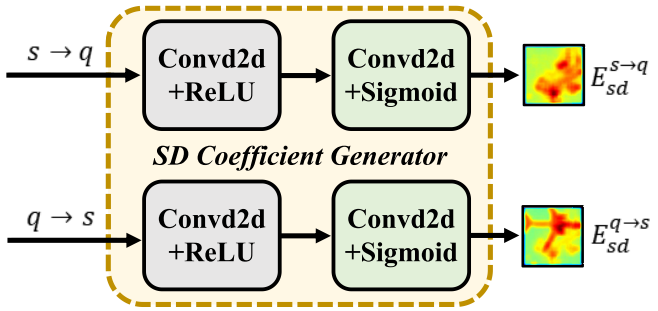


Fig. 4. Illustration of the SD coefficient generator.

where ϕ_s and ϕ_q represent the two-layer FC of two branches, respectively. Intuitively, only the channels with similar GSs of target regions in two branches could exhibit a significant activation in the mutual enhancement coefficient E_{gs}.

E. SD Mutual Enhancement Module

Due to the substantial differences in appearance, size, and distribution of similar target regions within remote sensing images, relying solely on GS enhancement is insufficient. In simpler terms, we need further take the SD mutual enhancement into account. Previous related works [13], [57] predominantly center on developing spatial interaction modules for mid-level features yet disregarding complementary information in high-level features. In this study, we integrate mid-level features with the task-agnostic prior masks (i.e., high-level features) generated from the DA module, constructing the SD mutual enhancement between the two branches.

Given the prototypes (i.e., v_s and v_q), we first expand them to fit the size of feature maps, $v_s^e \in \mathbb{R}^{C\times H\times W}$ and $v_q^e \in \mathbb{R}^{C\times H\times W}$. Regarding SD enhancement of the support-to-query, the expanded query prototype v_q^e is combined with the corresponding prior mask $\hat{M}_q$. Then, the enhancement coefficient is refined leveraging a smaller network $\mathcal{F}_{\text{sd}}^*$ comprising a 2-D convolution layer appended by a sigmoid activation function (see Fig. 4)

$$E_{\text{sd}}^{s\to q} = \mathcal{F}_{\text{sd}}^{s\to q}\left(\left(v_q^e \odot \hat{M}_q\right) \oplus v_s^e\right) \in \mathbb{R}^{H\times W} \tag{9}$$

where $\oplus$ denotes the element-wise sum. Similarly, the enhancement coefficient of the query-to-support can be computed as

$$E_{\text{sd}}^{q\to s} = \mathcal{F}_{\text{sd}}^{q\to s}\left(\left(v_s^e \odot \hat{M}_s\right) \oplus v_q^e\right) \in \mathbb{R}^{H\times W}. \tag{10}$$

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F. Mutual Fusion Decoder

Based on the two sets of coefficients (i.e., GS and SD mutual enhancements) produced from the GS and SD modules, we establish the holistic mutual enhancement to mitigate the issue of intra-class variances and background interference in a task-agnostic manner.

Concretely, both the prototypes and mid-level features are weighted by the two types of mutual enhancement coefficients, E_{gs}, $E_{\text{sd}}^{q\to s}$, and $E_{\text{sd}}^{s\to q}$

$$\begin{cases} \tilde{v}_s = v_s \odot E_{\text{gs}} \in \mathbb{R}^C \\ \tilde{v}_q = v_q \odot E_{\text{gs}} \in \mathbb{R}^C \\ \tilde{F}_s = F_s \odot E_{\text{gs}} \odot E_{\text{sd}}^{q\to s} \in \mathbb{R}^{C\times H\times W} \\ \tilde{F}_q = F_q \odot E_{\text{gs}} \odot E_{\text{sd}}^{s\to q} \in \mathbb{R}^{C\times H\times W}. \end{cases} \tag{11}$$

Then, the mutually enhanced mid-level features, $\tilde{F}_s$, $\tilde{F}_q$, prototypes, $\tilde{v}_s$, $\tilde{v}_q$, and prior masks, $\hat{M}_s$, $\hat{M}_q$ are all reshaped to the same spatial size and concatenated to construct support and query merged features, respectively,

$$\begin{cases} y_s^{\text{merg}} = \mathcal{C}(\tilde{F}_s, \mathcal{R}(\tilde{v}_q), \hat{M}_s) \in \mathbb{R}^{(2\times C+1)\times H\times W} \\ y_q^{\text{merg}} = \mathcal{C}(\tilde{F}_q, \mathcal{R}(\tilde{v}_s), \hat{M}_q) \in \mathbb{R}^{(2\times C+1)\times H\times W} \end{cases} \tag{12}$$

where $\mathcal{C}(\cdot,\cdot,\cdot)$ represents concatenating the channel dimension and $\mathcal{R}(\cdot)$ is the reshape function. Finally, the merged support feature y_s^{merg} and merged query feature y_q^{merg} are separately fed into two decoders to generate segmentation prediction for two branches

$$\begin{cases} M_s^{\text{pre}} = \mathcal{D}_s\left(y_s^{\text{merg}}\right) \in \mathbb{R}^{2\times H\times W} \\ M_q^{\text{pre}} = \mathcal{D}_q\left(y_q^{\text{merg}}\right) \in \mathbb{R}^{2\times H\times W} \end{cases} \tag{13}$$

where the two decoders $\mathcal{D}_s(\cdot)$ and $\mathcal{D}_q(\cdot)$ possess identical structures but have different parameters, which apply a convolution block followed by a classification head as in [49] to obtain the final prediction.

To guarantee the quality of HMRE, we introduce a bidirectional optimization paradigm to train the model. Given predicted mask M_q^{pre} and ground-truth query mask M_q, we adopt the BCE loss as our main loss

$$\mathcal{L}_{\text{seg}}^{s\to q} = \frac{1}{hw}\sum_{i=1}^{h}\sum_{j=1}^{w}\text{BCE}\left(M_q^{\text{pre}}(i,j), M_q(i,j)\right) \tag{14}$$

where h and w denote the height and width. Then we get the symmetric loss in the same paradigm

$$\mathcal{L}_{\text{seg}}^{q\to s} = \frac{1}{hw}\sum_{i=1}^{h}\sum_{j=1}^{w}\text{BCE}\left(M_s^{\text{pre}}(i,j), M_s(i,j)\right). \tag{15}$$

In summary, the total loss is constructed as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{seg}}^{s\to q} + \mu\mathcal{L}_{\text{seg}}^{q\to s} \tag{16}$$

where μ balances the contribution of each branch.

G. Extension to K-Shot Setting

In this case, there are K support images in each episode. Each one in the K annotated images contributes to segmenting the query image. Built upon this principle, we make some adjustments to certain steps of the model to fully utilize richer support representation. Note that our model does not need to be retrained under this setting. In other words, we leverage the trained model in the one-shot setting to perform support-to-query prediction during the inference stage.

During the DA phase, we yield K support prototypes with (1). Then, the cosine similarity between the query feature and the average of K support prototypes can be used to generate the activation map A_q. In addition, the K query prior masks generated from K support images are averaged to produce a more reliable query prior mask.

For the step of GS mutual enhancement, the query branch undergoes GS mutual enhancement with K support samples, respectively, resulting in K corresponding mutual enhancement coefficients, where the kth one can be similar to (8)

$$E_{\text{gs}}^k = \phi_s^k(v_s^k) \odot \phi_q(v_q) \in \mathbb{R}^C. \tag{17}$$

In the SD mutual enhancement part, the enhancement coefficient of the support-to-query can be modified as

$$E_{\text{sd}}^{s\to q} = \mathcal{F}_{\text{sd}}\left(\left(v_q^e \odot \hat{M}_q\right) \oplus \frac{1}{K}\sum_{k=1}^{K} v_s^{e|k}\right) \in \mathbb{R}^{H\times W}. \tag{18}$$

To further extend the mutual fusion decoder, both the prototypes and mid-level features are enhanced as follows:

$$\begin{cases} \tilde{v}_s^k = v_s^k \odot E_{\text{gs}}^k \in \mathbb{R}^C \\ \tilde{v}_q = v_q \odot \dfrac{1}{K}\displaystyle\sum_{k=1}^{K} E_{\text{gs}}^k \in \mathbb{R}^C \\ \tilde{F}_q = F_q \odot \dfrac{1}{K}\displaystyle\sum_{k=1}^{K} E_{\text{gs}}^k \odot E_{\text{sd}}^{s\to q} \in \mathbb{R}^{C\times H\times W}. \end{cases} \tag{19}$$

Correspondingly, the merged query feature is subsequently denoted as

$$y_q^{\text{merg}} = \mathcal{C}\left(\tilde{F}_q, \mathcal{R}\left(\frac{1}{K}\sum_{k=1}^{K}\tilde{v}_s^k\right), \hat{M}_q\right) \in \mathbb{R}^{(2\times C+1)\times H\times W}. \tag{20}$$

Finally, the merged query feature y_q^{merg} is fed into the decoder $\mathcal{D}_q$ to yield the predicted mask.

IV. Experiments

We first present the public dataset iSAID-5i, followed by the introduction of the newly created dataset Flood-3i. Next, the implementation details and the evaluation metric are demonstrated. Then, we report the performance of the HMRE approach and other advanced methods. Finally, ablation studies are conducted to show the impact of each component.

A. Datasets

1) iSAID-5i Dataset: We do extensive experiments on a public benchmark, named iSAID-5i [15], which originates from iSAID dataset [58]. It contains 18 076 samples for training and 6363 samples for testing. Each image is uniformly resized to a size of 256 × 256. Following the setting in [15], the distribution of class labels and quantities are shown in Table I.

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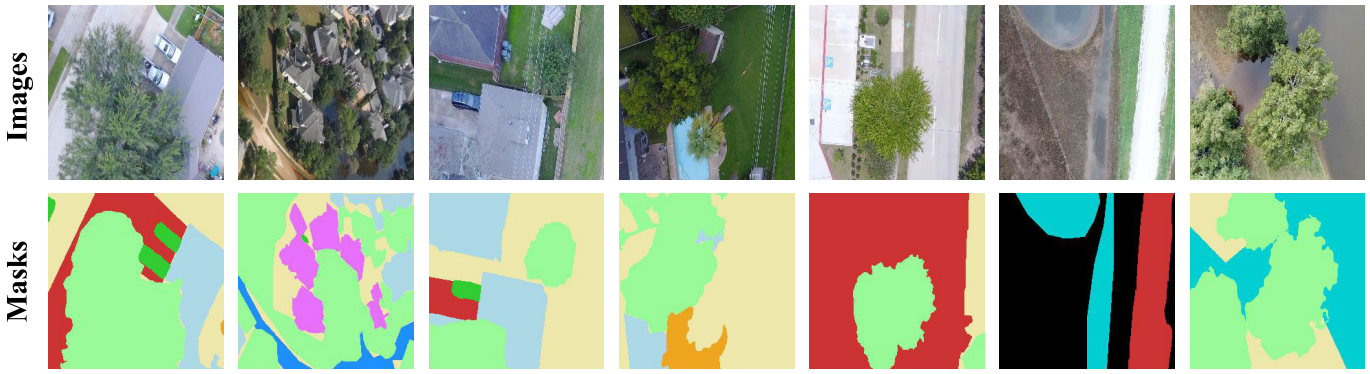


Fig. 5. Illustration of samples and corresponding masks in Flood-3i.

TABLE I
NUMBER OF IMAGES PER CATEGORY FOR THE iSAID-5i DATASET

Split ID	Category	Num of imgs in train set	Num of imgs in test set
Split0	ship	3820	1392
	storage tank	902	269
	baseball diamond	495	221
	tennis court	2485	767
	basketball court	598	160
Split1	ground track field	1405	517
	bridge	224	91
	large vehicle	2953	789
	small vehicle	2592	781
	helicopter	69	21
Split2	swimming pool	147	45
	roundabout	203	45
	soccer ball field	2000	682
	plane	1864	990
	harbor	3358	1361

2) Flood-3i Dataset: Presently, there is a scarcity of publicly available datasets for FSS research in remote sensing imagery, and they still possess certain limitations. The DLRSD [16] only involves 2100 RGB images, which diminishes the persuasiveness of algorithm evaluation. Despite the iSAID-5i dataset containing a substantial number of samples, each image has a limited variety of background classes. Instead, the FloodNet dataset [17], collected using unmanned aerial systems (UASs) for flood monitoring, offers 2343 high-resolution samples and encompasses nine object categories, presenting a compelling choice for conducting research in FSS tasks. Drawing upon this premise, we follow the setting in iSAID-5i [15] to create the Flood-3i dataset. Specifically, we uniformly crop the raw images and masks to a size of 256 × 256 to extend the available data. Ultimately, the complete Flood-3i dataset is created, consisting of 13 005 samples for training and 4050 samples for testing. Several examples from this dataset are illustrated in Fig. 5. For the nine classes in the Flood-3i dataset, models are evaluated using the cross-validation method by selecting three classes as test categories denoted as $\mathcal{D}_{\text{test}}$, while utilizing the remaining six classes as training categories designated as $\mathcal{D}_{\text{train}}$. The distribution of class splits is presented in Table II.

B. Implement Details

1) Experimental Setting: The implementation of our model is done in PyTorch [59] and it runs on an NVIDIA GeForce RTX 3090 GPU. The ResNet50 [31] or VGG16 [32] pretrained in ImageNet [60] are adopted as the encoder to extract features with freezing parameters. Images are sized to 256 × 256 to be forwarded through the FSS model. The proposed model is trained in an episodic scheme for 80 epochs, with a batch size of 16. The stochastic gradient descent (SGD) optimizer is employed in the training phase, and the initial learning rate is set to 0.0015. Additionally, a weight decay of 0.0001 is applied, and the learning rate is adjusted using the "poly" strategy with a power value of 0.9. The balance weight μ in the total loss is empirically set to 0.8 for optimal performance.

TABLE II
NUMBER OF IMAGES PER CATEGORY FOR THE FLOOD-3i DATASET

Split ID	Category	Num of imgs in train set	Num of imgs in test set
Split0	building flooded	869	304
	building non-flooded	1689	460
	road flooded	987	308
Split1	road non-flooded	2747	915
	water	3133	940
	tree	6457	2027
Split2	vehicle	171	46
	pool	277	76
	grass	5213	1480

2) Evaluation Metrics: We utilize the mean intersection over union (mIoU) as the evaluation metric. We define mIoU as the average over the set of all categories, i.e., $\text{mIoU} = (1/N)\sum_{i=1}^{N}\text{IoU}_i$, where N is the number of categories in each testing fold.

C. Comparison With State-of-the-Arts

We conduct a comparative analysis between HMRE and other advanced methods with the available source codes on the iSAID-5i and Flood-3i datasets. The mIoU performance of these advanced technologies including PANet [12], CANet [47], SD-AANet [50], PFENet [49], SDM [15], and SCL [52] are reported in Tables III–VI.

1) Results of mIoU on iSAID-5i: Tables III and IV show the one-shot and five-shot results on iSAID-5i. It is evident that our HMRE significantly surpasses other methods. Specifically, utilizing the ResNet50 backbone, the proposed method achieves a remarkable mean mIoU improvement of 6.27% in the one-shot setting and 2.34% in the five-shot setting

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TABLE III

COMPARISON WITH STATE-OF-THE-ARTS ON iSAID-5i DATASET IN MIOU UNDER THE ONE SHOT SETTING. RED/BLUE REPRESENTS THE FIRST-/SECOND-BEST PERFORMANCE

Backbone	Method	Split0						Split1						Split2						Mean
		C1	C2	C3	C4	C5	mean	C6	C7	C8	C9	C10	mean	C11	C12	C13	C14	C15	mean	
ResNet50	PANet	7.27	9.99	14.49	13.2	16.85	12.36	17.40	10.23	7.37	6.21	4.33	9.11	8.35	12.38	21.28	8.52	9.75	12.05	11.17
	CANet	7.25	27.25	43.32	6.17	9.99	18.80	6.16	17.81	28.58	8.84	16.73	15.62	17.73	56.97	13.82	27.41	13.06	25.79	20.07
	SD-AANet	25.21	39.85	43.60	21.69	22.27	30.52	35.18	47.28	23.89	25.47	14.75	29.31	9.46	27.38	20.51	12.50	22.45	18.46	26.10
	PFENet	2.98	6.30	33.29	34.49	16.71	18.75	8.68	8.67	34.62	18.19	16.08	17.24	28.58	24.47	7.37	39.76	10.31	22.09	19.36
	SDM	37.66	34.37	34.45	39.81	25.14	34.29	16.77	34.53	30.50	12.42	17.02	22.25	20.69	56.83	42.80	40.52	17.26	35.62	30.72
	HMRE(ours)	41.68	41.15	40.01	41.04	40.52	40.88	27.36	34.66	35.44	39.34	27.58	32.88	24.56	49.34	46.82	44.27	21.11	37.22	36.99
VGG16	PANet	13.85	17.78	19.78	17.8	17.94	17.43	22.41	11.75	11.75	6.36	4.88	11.43	12.96	19.11	23.49	11.87	12.33	15.95	14.94
	CANet	7.51	12.87	31.22	30.45	16.61	19.73	18.31	31.94	23.15	3.77	12.55	17.98	33.99	51.41	36.69	17.61	14.91	30.93	22.88
	SD-AANet	18.72	32.01	40.17	23.59	25.58	28.01	13.79	48.74	13.67	14.93	15.79	21.38	8.36	22.03	18.86	11.05	23.66	16.79	22.06
	PFENet	3.94	4.70	30.95	29.21	14.59	16.68	15.01	14.63	27.68	10.13	9.06	15.30	29.40	47.89	25.23	25.20	11.63	27.87	19.95
	SDM	26.55	30.57	33.01	35.06	21.03	29.24	20.54	28.89	32.45	9.62	12.59	20.80	30.79	43.52	49.47	27.51	22.37	34.73	28.26
	HMRE(ours)	30.25	36.16	41.88	43.37	30.74	36.48	31.52	36.66	35.10	18.38	19.74	28.28	29.74	45.09	49.22	30.82	21.93	35.36	33.37

TABLE IV

COMPARISON WITH STATE-OF-THE-ARTS ON iSAID-5i DATASET IN MIOU UNDER THE FIVE SHOT SETTING. RED/BLUE REPRESENTS THE FIRST-/SECOND-BEST PERFORMANCE

Backbone	Method	Split0						Split1						Split2						Mean
		C1	C2	C3	C4	C5	mean	C6	C7	C8	C9	C10	mean	C11	C12	C13	C14	C15	mean	
ResNet50	PANet	8.45	11.68	17.81	14.22	16.93	13.82	18.86	15.70	9.40	7.90	10.13	12.40	21.41	24.94	28.41	9.80	11.04	19.12	15.11
	CANet	32.47	6.49	39.8	25.97	14.59	23.86	25.01	25.27	32.06	1.31	9.08	18.54	26.99	73.84	33.83	20.69	4.69	32.00	24.8
	SD-AANet	40.53	37.08	46.10	29.67	37.31	38.14	38.48	49.39	26.21	30.27	16.85	32.24	9.84	28.05	21.31	19.79	22.54	20.31	30.23
	PFENet	10.13	9.48	30.71	31.23	16.31	19.57	11.53	14.15	36.07	15.58	14.82	18.43	38.52	20.90	20.41	38.66	12.22	26.14	21.38
	SDM	38.76	49.06	50.06	39.25	22.26	39.88	30.68	45.34	41.49	20.21	15.21	30.59	32.61	66.64	57.41	49.12	22.71	45.70	38.72
	HMRE(ours)	44.94	41.98	40.46	41.89	42.78	42.41	30.16	35.17	39.07	38.61	30.44	34.69	39.57	52.84	54.19	50.63	33.12	46.07	41.06
VGG16	PANet	13.40	17.79	19.04	18.11	20.15	17.70	22.97	16.18	15.65	9.37	8.73	14.58	31.65	21.63	25.36	12.70	12.15	20.70	17.66
	CANet	11.87	18.36	33.26	26.05	27.69	23.45	19.20	29.93	30.37	11.76	11.37	20.53	23.76	60.25	28.67	21.77	16.16	30.12	24.7
	SD-AANet	29.23	33.54	42.75	32.03	36.88	34.89	23.02	51.57	19.20	16.40	18.62	25.76	10.71	17.03	18.28	13.43	24.62	16.81	25.82
	PFENet	2.64	8.57	29.09	30.76	21.26	18.46	15.22	21.58	31.16	12.06	11.90	18.39	33.56	48.47	24.92	25.29	11.79	28.81	21.89
	SDM	30.81	38.59	37.34	42.58	32.35	36.33	32.32	39.71	37.23	17.73	12.93	27.98	39.20	66.78	39.06	46.17	20.72	42.39	35.57
	HMRE(ours)	31.74	36.18	42.79	42.59	33.34	37.33	33.02	37.51	35.63	20.09	20.14	29.28	42.72	47.15	47.22	45.93	33.78	43.36	36.66

compared to SDM, which is the second-best performing method. Simultaneously, with the VGG16 backbone, our one-shot and five-shot results respectively surpass the best competitor, i.e., SDM by 5.11% and 1.09% mean mIoU. For each split, the proposed model consistently attains the best performance and demonstrates a more significant advantage in scenarios with a very restricted number of samples (e.g., one-shot setting).

For a more thorough evaluation of performance across various categories, we also provide detailed results. It is clear that our proposed method excels in performance across almost all classes. Additionally, our method significantly boosts the segmentation performance of some challenging categories. In Table III, with the ResNet50 backbone, our model gets 15.38% mIoU improvement in the basketball court category (C5). With the VGG16 backbone, an enhancement of 9.11% is achieved over the second-best approach in the category of ground track field (C6). Under the five-shot setting (see Table IV), with the ResNet50 backbone, HMRE attains the optimal improvements, namely, 13.59%, in the helicopter category (C10). When employing the VGG16 backbone, the peak segmentation performance (progress of 9.16%) is obtained in the harbor category (C15). Meanwhile, the proposed HMRE exhibits suboptimal performance in the bridge class (C7). Upon thorough examination of the samples within this category, we have discerned that the uniform background and evenly distributed spatial elements might not be conducive to highlighting the advantages of our method.

2) Results of mIoU on Flood-3i: In Tables V and VI, we conduct a comparative analysis of our approach with other methods on the Flood-3i dataset. The HMRE gives a comprehensive optimal performance. For example, our approach outperforms the second-best methods (i.e., PFENet and SCL) by 5.02% and 6.22% mean mIoU under the one-shot and five-shot settings while employing the same backbone of ResNet50. With the VGG16 backbone, the proposed method obtains a mean mIoU 4.80% higher in the one-shot setting, and 3.25% in the five-shot setting. Besides, HMRE yields the best effectiveness in each split, which demonstrates its universality.

As for specific categories, our approach similarly shows its superiority. Particularly, although some methods (e.g., SD-AANet in split0 and split1, SCL in split0) excel in one category, they may experience a significant decline in other categories. In the aforementioned scenario, our approach exhibits superior performance while possessing a smaller degree of performance variance. In addition, for the more

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TABLE V

COMPARISON WITH THE STATE-OF-THE-ARTS ON FLOOD-3I DATASET IN MIOU UNDER THE ONE SHOT SETTING. RED/BLUE REPRESENTS THE FIRST-/SECOND-BEST PERFORMANCE

Backbone	Method	Split0				Split1				Split2				Mean
		C1	C2	C3	mean	C4	C5	C6	mean	C7	C8	C9	mean	
ResNet50	PANet	19.87	25.46	17.31	20.88	19.81	18.78	19.45	19.35	11.33	9.40	4.57	8.42	16.22
	CANet	22.89	17.54	19.82	20.08	25.29	18.62	14.05	19.32	9.22	6.86	6.28	7.45	15.62
	SD-AANet	23.70	52.93	9.34	28.66	46.79	5.32	30.09	27.4	15.87	4.62	4.83	8.44	21.5
	PFENet	44.42	47.20	30.87	40.83	44.52	38.52	29.88	37.64	6.40	14.17	2.65	7.74	28.74
	SCL	33.02	45.21	30.83	36.35	43.42	25.31	41.27	36.67	13.34	13.35	5.16	10.62	27.88
	HMRE(ours)	47.22	48.24	33.72	43.06	44.83	41.14	34.06	40.01	21.08	18.91	14.64	18.21	33.76
VGG16	PANet	16.02	27.12	12.38	18.51	17.30	14.49	21.48	17.76	12.73	6.25	4.37	7.78	14.68
	CANet	18.26	19.02	16.51	17.93	29.74	17.03	15.71	20.83	8.41	5.27	3.24	5.64	14.80
	SD-AANet	22.79	51.86	10.56	28.40	46.31	11.74	33.84	30.63	16.06	2.47	6.18	8.24	22.42
	PFENet	29.95	50.02	14.11	31.36	41.51	47.17	24.07	37.58	7.35	9.97	8.15	8.49	25.81
	SCL	31.32	57.17	10.37	32.95	46.37	27.14	34.61	36.04	11.31	12.51	4.95	9.59	26.19
	HMRE(ours)	30.40	37.80	37.05	35.08	41.86	47.94	29.21	39.67	20.73	18.32	15.59	18.21	30.99

TABLE VI

COMPARISON WITH STATE-OF-THE-ARTS ON FLOOD-3I DATASET IN MIOU UNDER THE FIVE SHOT SETTING. RED/BLUE REPRESENTS THE FIRST-/SECOND-BEST PERFORMANCE

Backbone	Method	Split0				Split1				Split2				Mean
		C1	C2	C3	mean	C4	C5	C6	mean	C7	C8	C9	mean	
ResNet50	PANet	25.59	28.77	19.43	24.60	24.29	19.90	23.46	22.55	12.26	9.52	8.93	10.24	19.13
	CANet	21.36	19.60	24.75	21.90	33.29	19.54	15.18	22.67	8.75	6.93	7.14	7.61	17.39
	SD-AANet	27.25	51.53	6.51	28.43	56.77	6.84	34.80	32.80	17.11	3.73	4.98	8.61	23.28
	PFENet	49.21	46.82	31.54	42.52	48.03	35.71	24.52	36.09	8.21	16.78	1.34	8.78	29.13
	SCL	35.48	46.64	32.95	38.36	44.12	25.04	44.85	38.00	15.73	12.06	7.21	11.67	29.34
	HMRE(ours)	50.61	49.68	36.14	45.48	45.79	41.95	36.66	41.47	22.67	18.96	17.55	19.73	35.56
VGG16	PANet	19.07	27.44	17.23	21.25	28.22	16.89	19.8	21.64	11.86	7.42	6.50	8.59	17.16
	CANet	19.25	21.83	17.30	19.46	36.74	15.43	14.97	22.38	11.40	7.73	8.69	9.27	17.04
	SD-AANet	16.75	53.77	4.92	25.15	54.15	12.85	40.19	35.73	16.71	2.40	5.75	8.29	23.06
	PFENet	33.28	51.82	7.28	30.79	47.91	35.71	24.52	36.05	5.77	8.68	7.29	7.25	24.70
	SCL	35.24	52.83	9.23	32.43	45.51	19.66	44.77	36.65	13.22	8.00	4.31	8.51	25.86
	HMRE(ours)	30.25	36.76	35.59	34.20	39.27	45.19	31.68	38.71	16.92	14.89	11.43	14.41	29.11

challenging category fold split2 characterized by an imbalanced number of samples, the proposed approach reaps tremendous performance improvements. We believe that this can be attributed to the successful implementation of the HMRE, which enables the model to mine valuable information from support images.

3) Segmentation Examples: To facilitate more in-depth analysis and comprehension of the proposed model, we showcase several segmentation examples by the baseline (i.e., PFENet [49]) method and our HMRE approach, as shown in Fig. 6. It can be seen that HMRE presents significantly more accurate prediction masks. For instance, when handling the minuscule masked objects in the third column (belonging to the "vehicle" category), our approach can still generate satisfactory results for the query image.

D. Ablation Studies

To examine the effect of each factor on segmentation accomplishment, a series of ablation studies are conducted with ResNet50 backbone and we set PFENet [49] as the baseline.

1) Components Analysis: The proposed HMRE contains two major components, i.e., GS and SD mutual enhancement modules. Note that we additionally incorporated a GS mutual enhancement strategy in CRNet [14] to probe the superiority of the devised GS module, which this article refers to as GS-CR. Under the Flood-3i dataset, Table VII showcases the effectiveness of each component. Compared with the baseline, the utilization of GS, GS-CR, and SD modules all contribute to a substantial improvement in the segmentation consequence. By combining the GS and SD modules, HMRE achieves the best performance. Moreover, the GS module exhibits a greater performance gain for the model compared to the

TABLE VII

ABLATION STUDIES OF MAJOR MODEL COMPONENTS USING MIOU (%) ON FLOOD-3I DATASET

Shot	Baseline	GS-CR	GS	SD	Split0	Split1	Split2	Mean
1	✔				40.83	37.64	7.74	28.74
1	✔	✔			41.16	38.19	10.90	30.08
1	✔		✔		42.27	39.33	14.46	32.02
1	✔			✔	41.25	38.89	15.03	31.72
1	✔	✔		✔	42.34	39.48	16.73	32.85
1	✔		✔	✔	**43.06**	**40.01**	**18.21**	**33.76**
5	✔				42.52	36.09	8.78	29.13
5	✔	✔			43.66	38.39	12.82	31.62
5	✔		✔		44.52	40.80	16.29	33.87
5	✔			✔	44.30	38.59	15.70	32.86
5	✔	✔		✔	45.32	39.87	17.84	34.34
5	✔		✔	✔	**45.48**	**41.47**	**19.73**	**35.56**

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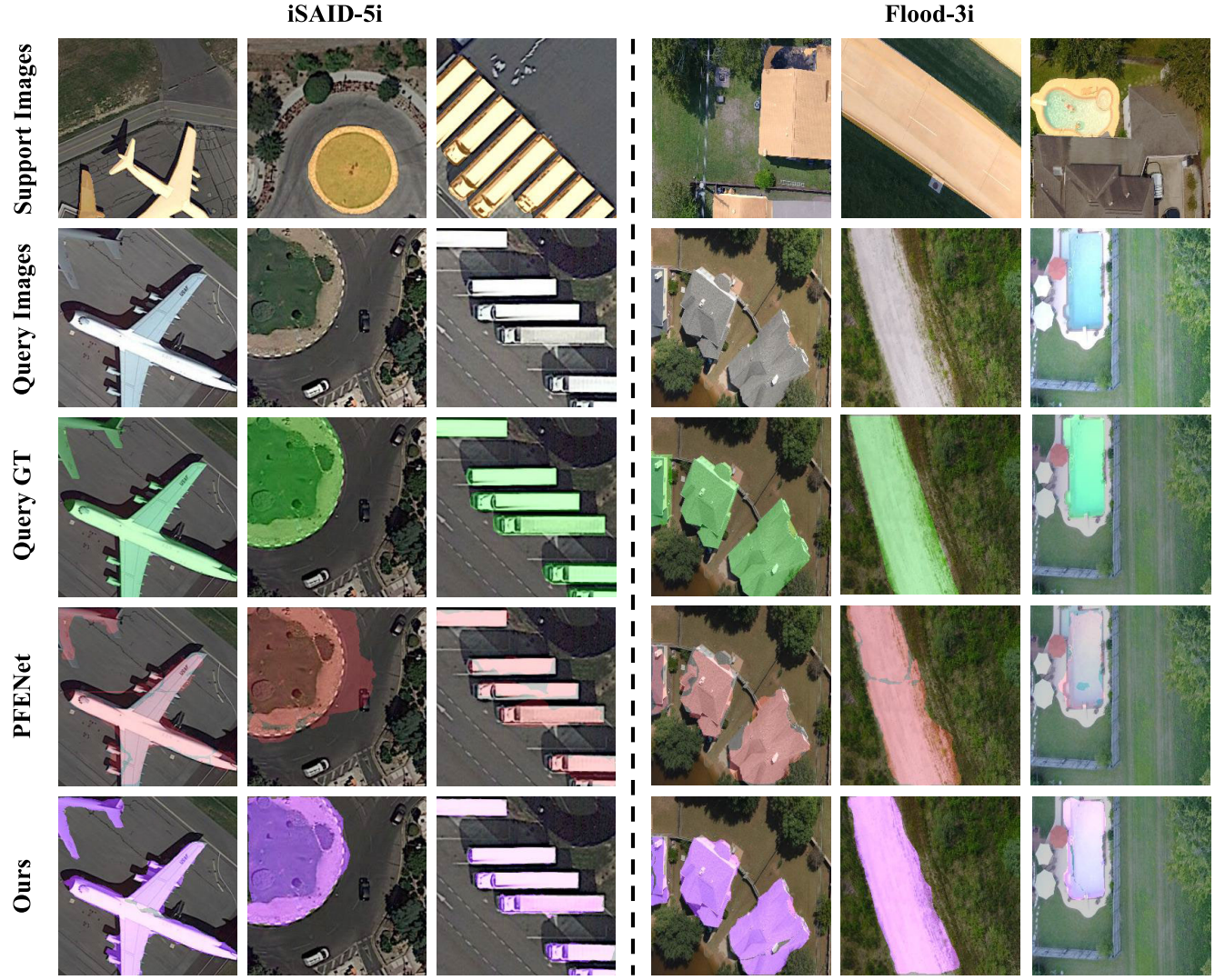


Fig. 6. Segmentation examples of HMRE and baseline method (PFENet) under the one-shot setting. The left panel is from iSAID-5i, and the right one is from Flood-3i. Images within the same column correspond to a certain object category. The specific target regions include planes, roundabouts, vehicles, building non-flooded, road non-flooded, and pools.

GS-CR strategy. This is attributed to its effective suppression of background interference in remote sensing images.

2) Effect of the Mutual Fusion Schemes: As described in Section III-F, by leveraging the two types of mutual enhancement coefficients, we propose to establish mutual fusion by weighting the prototypes and mid-level features. To study the influence of different mutual fusion schemes, we qualitatively visualize the activation maps, as illustrated in Fig. 7. From left to right, each column represents: support images with annotation masks [Fig. 7(a)], query images with the GT [Fig. 7(b)], baseline results [Fig. 7(c)], the results of weighting prototypes [Fig. 7(d)], the results of weighting mid-level features [Fig. 7(e)], and the results of both weighting prototypes and mid-level features (ours) [Fig. 7(f)].

It can be observed that the baseline method erroneously activates certain non-target regions, whereas the GS module effectively suppresses non-target category interference within the background. Furthermore, the SD module is beneficial for mining more edge and detail features, particularly in cases where the spatial distribution of the targets is densely concentrated. When incorporating both the SD and GS modules, the model enables more accurate activation of the target regions within the query samples.

3) Effect of the Hyperparameter μ for Balancing the Bidirectional Optimization: We undertake an analysis to explore the influence of the hyperparameter μ, which adjusts the contribution of optimization in two directions. Concretely, the experiments are conducted on iSAID-5i and Flood-3i datasets with ResNet50 backbone. We regulate μ in the interval $[0, 2]$, as shown in Fig. 8. When μ is set as 0, it is equivalent to the model undergoing only unidirectional optimization from support to query. This leads to the loss of effective supervision for mutual representation enhancement between the two branches, thereby resulting in a decline in model performance. As μ increases, the segmentation accuracy continuously improves due to the gradually strengthening supervision. For the two datasets and all settings, HMRE achieves the peak performance when μ is set to 0.8. Nevertheless, further increasing μ results in a decrease in the contribution of support-to-query

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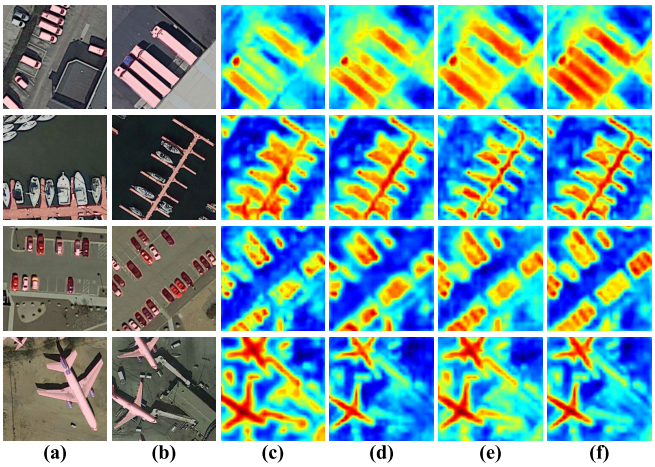


Fig. 7. Activation maps with different mutual fusion schemes under the one-shot setting. The target classes of the top to bottom rows are "large vehicle," "harbor," "small vehicle," and "plane," respectively. (a) Support images with annotation masks. (b) Query images with GTs. (c) Baseline results. (d) Weighting prototypes. (e) Weighting mid-level features. (f) Weighting both prototypes and mid-level features (ours).

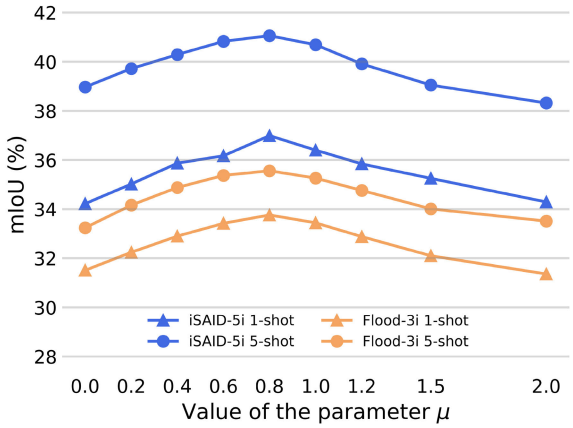


Fig. 8. Impact of the parameter μ for the FSS mIoU.

direction optimization, reducing the segmentation accuracy of the model for query samples.

E. Extended Experiments

1) Model Discriminability: We randomly select 1000 episodes from nine categories of the Flood-3i dataset to create a confusion matrix, as depicted in Fig. 9. It is evident that the baseline method demonstrates lower discriminability among regions of similar categories (e.g., c1 and c2, c3 and c4), making it susceptible to erroneous predictions. In comparison, the proposed HMRE effectively mitigates semantic confusion and demonstrates superior segmentation performance.

2) Cross-Set FSS: Considering a scenario that is more common and challenging in practical applications, we further perform the Cross-set FSS experiment. We utilize split1 and split2 from the iSAID5-i training set as the base (seen) data, and split0 from the Flood-3i test set as the novel (unseen) data, refer to Tables I and II for the distribution of specific target categories. The results presented in Table VIII clearly

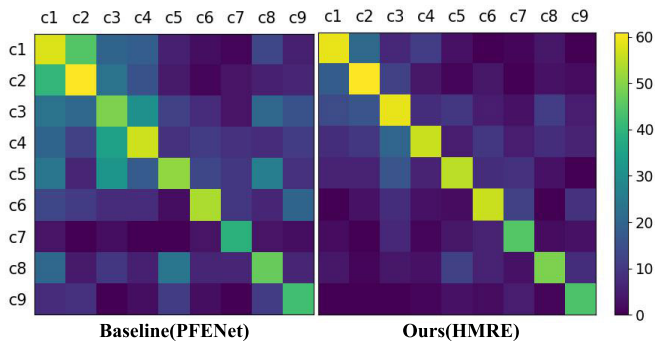


Fig. 9. Confusion matrix for the target categories on Flood-3i.

TABLE VIII
EXTENDED EXPERIMENT OF CROSS-SET FSS ON iSAID-5i → FLOOD-3i UNDER THE ONE SHOT AND FIVE SHOT SETTINGS. ALL DATA REPRESENT THE mIoU RESULTS OBTAINED FROM 2000 EPISODES

Backbone	Method	iSAID-5i → Flood-3i	
		1-shot	5-shot
ResNet50	Baseline (PFENet)	29.97	31.93
	Ours (HMRE)	34.63	36.29
VGG16	Baseline (PFENet)	22.77	25.71
	Ours (HMRE)	30.18	33.24

demonstrate that the proposed HMRE approach outperforms the baseline method significantly in all settings. This indicates that our model exhibits superior generalization capabilities and holds promising potential for practical applications.

V. CONCLUSION

This article proposes the HMRE approach for mitigating the intra-class variances and background interference in FSS, which constructs mutual representation enhancement between two branches. Particularly, the DA module establishes information symmetry within the dual-branch structure. Then, we decouple the holistic mutual enhancement as the GS and SD mutual enhancement modules. Ablation experiments were organized to qualitatively and quantitatively probe the effectiveness of this strategy. To optimize the process of mutual enhancement, the devised mutual fusion decoder adopts a bidirectional prediction paradigm. Moreover, we curate a new dataset called Flood-3i, which aims to facilitate the research of FSS. The comprehensive experiments conducted indicate the efficacy of our method, establishing it as the new state-of-the-art.

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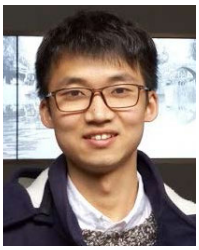
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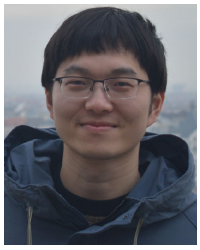
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